Victoria You

(615) 484-1850 | vyou2@illinois.edu

https://www.linkedin.com/in/victoria-you/

EDUCATION

University of Illinois Urbana-Champaign *Expected May 2028*

3.95/4.00; B.S Computational Physics; B.S Mechanical Engineering; Minor in Semiconductor Engineering

- Dean's List; James Scholar Honors; Pi Tau Sigma; ISUR Research Scholar; Engineering Visionary Scholarship
- Relevant Coursework: Semiconductor Devices; Robotic Manipulators; Design for Manufacturability; Fluid Dynamics; Classical Mechanics; Thermodynamics; Solid Mechanics; Electrical and Electronic Circuits; Computer Science I & II

PROFESSIONAL EXPERIENCE

UIUC Mechanical Science & Engineering Jan 2026- *Present*

Web Content Developer *Champaign, IL*

- Developing and maintaining online reference pages for ME 270: Design for Manufacturing coursework and materials
- Typesetting lecture slides as technical documents in LaTeX and generating supplemental examples and figures
- Managing version control and collaborative workflows with GitHub while optimizing functionality and accessibility

MIT Lincoln Laboratory Sep 2025- Jan 2026

Microfabrication Process Engineering Co-Op *Lexington, MA*

- Increased multi-wafer uniformity of HZO Atomic Layer Deposition films by 88% with iterative recipe optimization
- Refined stability and conditioning for metals E-beam Evaporation, improving Hf deposition conductivity by 49%
- Developed Selective Si Epitaxial Growth processes for in-house fabrication of RSD transistors and sub 90nm CMOS

Cygnus Photonics, Inc. Feb 2025-Aug 2025

Research & Development Engineering Intern *Champaign, IL*

- Manufactured UV/VUV exposure systems for customer research and performed post-use troubleshooting and repair
- Facilitated incoming quality control to drive design revisions; created 25+ technical part drawings for manufacturing
- Designed and iterated modular experimentation units from 3D-printed and laser cut prototypes to machined products
- Reprogrammed a 3D-Printing Gantry using LabVIEW to determine uniformity for lamp area increases of up to 500%

Nashville Motorcycle Repair May 2024-Aug 2024

Restoration Mechanic Intern *Nashville, TN*

- Fully restored non-running motorcycles for resale: Honda Z50 Minibike; 1982 Suzuki GS1100EZ
- Partially restored motorcycles in poor condition for resale: 1996 Moto Guzzi Sport 1000; Suzuki LS650 Savage Ryca
- Rebuilt and serviced all customer Ural Motorcycle wheels affected by the 2011-2017 Ural Factory Rim Recalls

Vanderbilt University Advanced Robotics & Mechanism Applications Lab Jun 2023-Aug 2023

Research Intern *Nashville, TN*

- Developed a MATLAB simulation to visualize a moving variable continuum link using a resolved rates algorithm
- Integrated modules to calculate and graph robot inverse kinematics and end-effector speeds in real time
- Designed a 3D Printed continuum model powered by Arduino, implementing resolved rates for joystick control

PROJECTS

Variable Filament Extruder, *Clemon Research Group* Jun 2025- *Present*

- Designing a direct-drive 3D-printer extruder to process non-uniform recycled filament without loss in print quality
- Using MATLAB to optimize extruder design for different filament distributions and manufacturing tolerance analysis

NovoPrint Multi-Extruder Arm, *American Society of Mechanical Engineers @Illinois* Jan 2025-*Present*

- Leading a team of 15+ undergraduates to design and manufacture two 6-Axis robot arms for non-planar 3D-printing
- Path planning and simulating motion using ROS Rviz, controlling via CAN bus for simultaneous multi-arm printing
- Awarded 3rd Place Distinguished Technology Award at the 2025 Illinois Engineering Open House

3D Printing Program for Homeschoolers, *Brentwood Public Library* Mar 2024-Aug 2024

- Established an Introduction to 3D Printing Class for homeschoolers and younger students without 3D printer access
- Trained library staff to troubleshoot, diagnose, and repair common FDM printer issues

VEX Robotics Competitions, *Ravenwood Robotics Club* Sep 2021-Apr 2024

- Spearheaded design and build processes to win the TN State Championship Build Award for two consecutive years
- Coordinated the documentation of the team Engineering Design Process in a 700-page engineering notebook
- Mentored multiple teams to qualify for VEX Robotics WORLDS and coordinated logistics of the multi-day trip

SKILLS & INTERESTS

- Software: MATLAB, JMP, Java, Python, ROS, C++, Arduino, SolidWorks, Autodesk Inventor, Autodesk Fusion 360
- Manufacturing: Rapid Prototyping, FDM & SLA 3D Printing, Laser Cutting, Soldering, GD&T, Microfabrication
- Hobbies & Interests: Varsity Illini Rowing, Motorcycles, Snowboarding, Origami Paper Cranes